

ESP32 RTC BASED CAMERA

Using Motion Sensor

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In the previous version of ESP32 cam project published in February issue you could live stream the video and access the video over Wi-Fi. Carrying that forward, in this project the ESP32 cam attached with RTC can take pictures based on time and save them on an SD card attached to the board. The design can be further developed to capture a video picture only when some motion is detected, so that it can act as a wireless security camera.

The author's prototype is shown in Fig. 1 while the components required for the project are listed under Bill of Material table. The circuit diagram for the ESP32 RTC based camera is shown in Fig. 2.

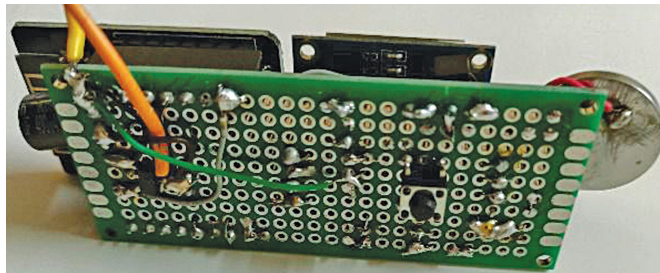


Fig. 1: Author's prototype soldered on board

The circuit is built around voltage regulator LM1117, an OLED, and ESP32 cam. For programming the ESP32 you need the USB-to-serial programmer shown in Fig. 3.

Software

The ESP32 camera does not come with built-in USB-to-serial interface. Therefore, to upload the sketch, one needs a USB-to-serial programmer, which is shown on the right side in Fig. 3. The GPIO 0 pin is to be pulled to ground during uploading mode and the reset button of the ESP32 is to be pressed during first few seconds of uploading and then released.

For capturing the video based on time, the ESP32 camera web server code is customised and the I2C based RTC library is used to fetch the data and time from I2C based RTC

module. The camera and the SD card engage most of the GPIO pins, leaving very few GPIO pins for work.

Pin 4 is used for flashlight, deploying the built-in ESP32 super bright LED as an extra light source for the camera. Pin 13 is used as the trigger pin for taking pictures. The program stamps the time on the file name and then stores serially in the SD card.

Normally, the ESP32 goes into deep sleep mode after taking a picture. Pin 13 is used as a trigger to

BILL OF MATERIAL

Components	Quantity
ESP32 with camera (MOD1)	1
OLED display, 2.44cm (0.96-inch) (MOD2)	1
I2C RTC – DS3231/DS1307	1
USB to serial FTDI chip	1
LM1117 3.3V regulator	1
PIR sensor	1
Optocoupler	1
Transistor BC547	1
10k-ohm resistor and 100 μ F capacitor	2 and 1

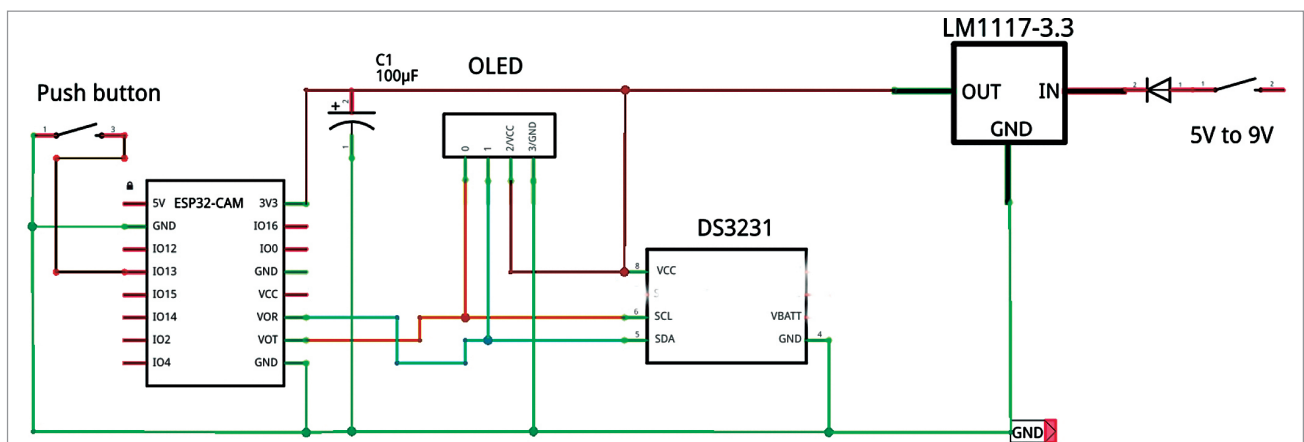


Fig. 2: ESP32 RTC based camera circuit